Additional arithmetic hypotheses and polynomial definitions of the integers

9 November 2026

Abstract

We distinguish height inequalities that shorten a polynomial definition from prime-existence statements and from conjectures about rational points. A weak *abc* inequality with exponent 121/100 suffices for the thirteen-coordinate construction; consequently the ordinary discriminant–conductor form of Szpiro's conjecture suffices. The truncated Vojta conjecture supplies the stronger ordinary *abc* inequality. We prove precise limitations of two other proposed reductions: distinct prime assignments do not control the squarefree part of a product, and a polynomial injection does not by itself give a finite algebraic encoding of several free coordinates. Finally, a topological argument explains the incompatibility between Mazur's connected-component conjecture and an existential definition of the integers. No reduction in the number of coordinates from the Riemann hypothesis or Grimm's conjecture is asserted.

1 Definitions and the required height estimates

All coordinate counts include the input. A polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$ defines $\mathbb{Z}$ in $\mathbb{Q}$ if

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^m \quad F(x, \mathbf{y}) = 0.$$

For a nonzero integer n, put

$$\operatorname{rad}(n) = \prod_{p \mid n} p, \qquad \operatorname{sf}(n) = \prod_{\substack{p \mid n \\ v_p(n) \text{ odd}}} p.$$

For $t = u/v$ in lowest terms with $v > 0$, let $h(t) = \log \max(|u|, v)$. If (w, u, v) is a primitive common integer representative of $(1, a, b)$ with $w > 0$, define $h(1 : a : b) = \log \max(w, |u|, |v|)$.

Definition 1. *For $\kappa > 1$, the weak abc inequality with exponent κ is the assertion that a constant $C_\kappa > 0$ exists such that*

$$\max(|a|, |b|, |c|) \le C_\kappa \operatorname{rad}(abc)^\kappa$$

for every primitive nonzero integer triple $a + b = c$.

The ordinary rational *abc* conjecture asserts this for every $\kappa > 1$. The following result uses only one exponent.

Theorem 2. *Assume the weak abc inequality with exponent* 121/100 *and the elliptic group, pole obstruction and realization, and simultaneous prime-pattern inputs specified in [1]. There is a computable coefficient family in thirteen total coordinates whose sufficiently large fixed odd input shifts define $\mathbb{Z}$ in $\mathbb{Q}$. A sufficient numerical shift is not determined. The same conclusion follows under the ordinary Szpiro conjecture and the stated inherited inputs.*

Proof. We describe the numerical inequalities that make the extra hypothesis sufficient; the complete rational source and polynomial conversion are given in [1]. The rational map

$$f(t) = \frac{t(t+2)^3}{(2t+1)^3}, \qquad f(t) - 1 = \frac{(t-1)^3(t+1)}{(2t+1)^3}$$

has degree four. Its inverse images of $0, 1, \infty$ are the six rational points $0, -2, -1/2, 1, -1, \infty$. Applying the weak inequality to its numerator, denominator, and their difference gives

$$\log M(t) \geq (8/\kappa - 6)h(t) - O(1),$$

where M is the product of primes having odd contact valuation, outside a fixed exceptional set. To see the coefficient, the radical of the associated degree-six binary form is at least a constant times $\exp(4h(t)/\kappa)$, whereas $\mathrm{rad}(n)^2 \leq |n|\,\mathrm{sf}(n)$ and the form has size $O(\exp(6h(t)))$. The fixed change $t = 2 + 1/\lambda$ makes all six contacts finite.

The same weak inequality, applied to coprime eighth powers, gives

$$h(1 : \beta : T) \leq \frac{121}{437} h(\beta^8 + T^8) + O(1). \tag{1}$$

The common-denominator calculation proving (1) is included in [1]; its general coefficient is $(d/\kappa - 3)^{-1}$ for even $d > 3\kappa$. For the two arguments λ and $(\beta^8 + T^8)\lambda$, the contact bound and the local congruences therefore give

$$h(1 : \beta : T) \leq \frac{14641}{16169} \log N + O(1) \tag{2}$$

whenever $N = |\operatorname{num}(\beta - n)| \neq 0$. Here $\beta = 2X^2 + B$, the elliptic index n is an integer, and the one-point height theorem gives $n^2 \leq Ah(T) + J$ with fixed constants. Since $\log N \leq h(\beta) + \log(1 + |n|)$, (2) bounds both n and $h(\beta)$ independently of the fixed odd shift B. Choosing B larger than the exponential of the bound for $h(\beta)$ forces $\beta = n$. The elementary denominator calculation for $2X^2 + B$ then gives $X \in \mathbb{Z}$. Completeness and the thirteen-coordinate count follow from the explicit source in [1].

The semistable case of the ordinary Szpiro conjecture is the assertion $|\Delta_E| \leq C_\epsilon N_E^{6+\epsilon}$ for semistable elliptic curves over $\mathbb{Q}$, with minimal discriminant Δ_E and conductor N_E. Its implication of the weak *abc* inequality with every exponent $6/5 + \eta$ is established in [2, §I.3]. Take $\eta = 1/100$. □

This implication uses the discriminant–conductor conjecture itself. It does not identify that conjecture with the stronger modified Szpiro statement about $\max(|c_4|^3, |c_6|^2)$. The modified statement is equivalent to ordinary *abc*; see [2, §I.3]. The thirteen-coordinate conclusion does not remove the pole or elliptic realization hypotheses.

2 The truncated Vojta conjecture

Let X be smooth and projective over $\mathbb{Q}$, let D be a reduced divisor with normal crossings, and let A be ample. The rational-point case of the truncated Vojta conjecture asserts, outside a proper closed exceptional subset, that

$$N^{(1)}(D, P) \geq h_{K_X + D}(P) - \epsilon h_A(P) - O(1). \tag{3}$$

The counting function records a prime once when the reduction of P meets D. This is the specialization of the conjecture in [3, Conjecture 2.3].

Proposition 3. *The case of (3) for $X = \mathbb{P}^1$, $D = [0] + [1] + [\infty]$, and $A = [\infty]$ implies the ordinary rational abc conjecture.*

Proof. Write $P = a/c$ with $a + b = c$ primitive and nonzero. At finite primes the truncated counting function is $\log \operatorname{rad}(abc)$. The divisor $K_X + D$ has degree one, so its height is $h(P) + O(1)$. Thus (3) gives

$$(1 - \epsilon)h(P) \leq \log \operatorname{rad}(abc) + O(1).$$

The difference between $h(P)$ and $\log \max(|a|, |b|, |c|)$ is bounded by $\log 2$. A proper closed subset of $\mathbb{P}^1$ is finite, so its remaining rational points can be absorbed into the constant. Taking $\epsilon = \eta/(1 + \eta)$ gives exponent $1 + \eta$. □

Consequently this Vojta hypothesis supplies every additional *abc* input of the thirteen-coordinate and conditional quartic constructions. A lower coordinate count does not follow merely from replacing *abc* by this stronger conjecture.

3 Prime assignments and odd valuations

Grimm's conjecture asserts that a finite block of consecutive composite positive integers admits distinct prime divisors, one for each member. Its formulation and its relation to prime gaps are discussed in [4]. The next elementary example isolates a difficulty in using such assignments as a height estimate.

Proposition 4. *There are arbitrarily large positive integers n such that n and $n + 1$ are both composite and*

$$\operatorname{sf}(n(n + 1)) = 2.$$

Each such pair admits distinct prime divisors.

Proof. Define positive integers a_k, b_k by

$$(17 + 6\sqrt{8})^k = a_k + b_k\sqrt{8}, \qquad k \geq 1.$$

Their norm identity is $a_k^2 - 8b_k^2 = 1$, and a_k is odd. For $n_k = (a_k - 1)/2$ this gives $n_k(n_k + 1) = 2b_k^2$. Multiplication by $17 + 6\sqrt{8}$ acts by the integral matrix

$$\begin{pmatrix} 17 & 48 \\ 6 & 17 \end{pmatrix},$$

whose determinant is one. Its image in $\mathrm{GL}_2(\mathbb{Z}/144\mathbb{Z})$ has some finite order L. For $k \equiv 1 \pmod{L}$ we therefore have $a_k \equiv 17 \pmod{144}$, whence $n_k \equiv 8 \pmod{72}$. In particular $8 \mid n_k$ and $9 \mid n_k + 1$; both are composite. These n_k tend to infinity. Their consecutive members are coprime, so any prime divisor of one differs from every prime divisor of the other. □

Proposition 4 concerns the information supplied by a prime assignment. It is not an independence theorem between conjectures. In particular, it shows why assigning different primes to factors cannot be substituted for a lower bound on primes occurring to odd order.

The Riemann hypothesis concerns the zeros of the Riemann zeta function; its generalized form concerns zeros of the appropriate L-functions. Their usual prime-counting consequences concern the distribution of primes, as illustrated by Schoenfeld's conditional estimates [5]. Such a counting estimate does not supply the uniform height inequalities (1)–(2). No implication from either hypothesis to those inequalities, or to a smaller defining polynomial, is used here. Similarly, Schinzel-type prime-value hypotheses could alter a completeness argument only after the relevant simultaneous polynomial values and admissibility conditions have been specified. They do not by themselves replace the soundness argument.

4 Polynomial injections and finite algebraic encodings

The Bombieri–Lang conjecture predicts that the rational points of a projective variety of general type are not Zariski dense. For surfaces, Poonen proves that this conjecture implies the existence of a polynomial inducing an injection $\mathbb{Q}^2 \to \mathbb{Q}$ [6, Theorem 1.1]. His Remark 1.7 records a different sufficient hypothesis: the four-term generalized *abc* conjecture. It is essential to distinguish that conjecture from ordinary, three-term *abc*.

Proposition 5. *Let k be a characteristic-zero field and let U be a nonempty open subset of $\mathbb{A}^r_k$, with $r > 1$. No rational function $j : U \dashrightarrow \mathbb{A}^1_k$ is generically finite onto its image.*

Proof. If j is constant, its only fiber has dimension r. Otherwise $k(j)$ has transcendence degree one over k, whereas $k(U) = k(t_1, \dots, t_r)$ has transcendence degree r. Thus $\operatorname{trdeg}_{k(j)} k(U) = r - 1 > 0$, which is incompatible with a finite function-field extension. □

A polynomial may nevertheless be injective on k-rational points, since geometric and rational fibers are different. If j is injective on $\mathbb{Q}^r$ and a relation is $F(x, \mathbf{y}) = 0$, encoding its rational witnesses by $z = j(\mathbf{y})$ gives the exact reformulation

$$\exists \mathbf{y} \in \mathbb{Q}^r : F(x, \mathbf{y}) = 0, \quad z = j(\mathbf{y}). \tag{4}$$

Injectivity proves uniqueness of a rational inverse when one exists; it does not supply its existence or remove the quantifiers in (4). Proposition 5 also excludes replacing that inverse on an open set by a finite algebraic extension, the setting required by the one-coordinate separation theorem. This is a limitation of that proposed encoding. It does not rule out a different use of polynomial injections in a Diophantine construction.

5 A conjecture incompatible with the projection

Mazur's connected-component conjecture asserts that for every variety $V/\mathbb{Q}$, the closure of $V(\mathbb{Q})$ in $V(\mathbb{R})$ has finitely many connected components. The implication below is classical; see [7, §§2–3].

Proposition 6. *If Mazur's connected-component conjecture holds, $\mathbb{Z}$ is not Diophantine in $\mathbb{Q}$.*

Proof. Suppose a polynomial $F(x, \mathbf{y})$ defines $\mathbb{Z}$, and let V be its affine zero set. Let C be the real topological closure of $V(\mathbb{Q})$ in $V(\mathbb{R})$. The coordinate projection $\pi : V(\mathbb{R}) \to \mathbb{R}$ is continuous. Since $\pi(V(\mathbb{Q})) = \mathbb{Z}$ and $\mathbb{Z}$ is closed, $\pi(C) \subseteq \mathbb{Z}$. Conversely every integer occurs at a rational point, so $\pi(C) = \mathbb{Z}$. Every connected component of C has a connected image in the discrete set $\mathbb{Z}$, hence a one-point image. Finitely many components would give a finite image, a contradiction. If the conjecture is formulated only for irreducible varieties, apply it to the finitely many irreducible components of V. □

Thus this topological conjecture cannot be adjoined to arithmetic premises already implying an existential definition of $\mathbb{Z}$ as a consistent method of reducing the number of witnesses. It is distinct from the Bombieri–Lang statement used above.

References

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